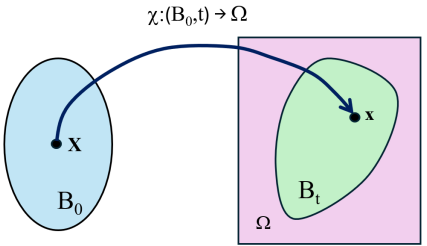
WHY COMBINE IMMERSED BOUNDARY AND FINITE ELEMENT METHODS?

The mechanics, coupling, and tradeoffs of IBFE

Laura A. Miller

Department of Mathematics, University of Arizona

Fluid Dynamics and IBM Theory • Module 4



Schematic: instructor's course notes.

Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.

ROADMAP

1. Why a fixed fluid grid helps with large structural motion
2. How finite elements represent the solid mechanics
3. How integral operators exchange velocity and force
4. Which approximations and tradeoffs matter in practice

> IBFE keeps the immersed interface treatment and uses a variational finite element model for solid mechanics.

Unless noted otherwise, the formulation and terminology in this deck are adapted from Griffith & Luo (2017), which builds on the continuous IB formulation of Boffi, Gastaldi, Heltai & Peskin (2008); full references are on the final slide.

THE FLUID–STRUCTURE INTERACTION CHALLENGE

Fluid description

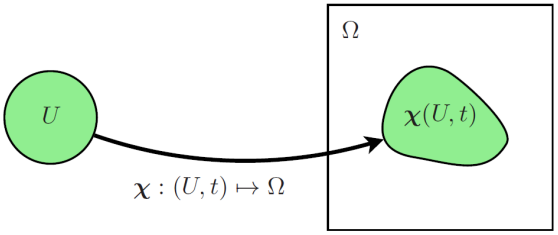
- A fixed Eulerian grid supports efficient incompressible-flow solvers.
- Velocity and pressure live at spatial locations.
- The grid should remain well shaped as the structure moves.

Solid description

- A material mesh follows deformation and rotation.
- Constitutive laws depend on strain relative to a reference state.
- Large motion can severely distort a body-fitted fluid mesh.

> The numerical method must enforce no slip and transmit elastic force without requiring the two meshes to conform.

REFERENCE AND CURRENT CONFIGURATIONS



Reference configuration B_0, current configuration B_t, and the motion $\boldsymbol{\chi}$. Figure 1 of Griffith & Luo (2017), *Int. J. Numer. Methods Biomed. Eng.* 33:e2888. NB: the figure labels the reference domain U; in this course it is B_0 (and their material point $\boldsymbol{s}$ is our $\boldsymbol{X}$). The title-slide schematic shows the same picture in course notation.

$$\boldsymbol{\chi} : B_0 \times [0,T] \longrightarrow \Omega$$

$$\boldsymbol{x} = \boldsymbol{\chi}(\boldsymbol{X}, t)$$

- B_0: reference solid domain
- $B_t = \boldsymbol{\chi}(B_0, t)$: current solid region
- Ω: fixed Eulerian domain
- $\boldsymbol{X}$: material coordinate
- $\boldsymbol{x}$: spatial coordinate

Notation change from Module 1: there the current position of material point q was written $\boldsymbol{X}(q,t)$. From here on we follow Griffith & Luo: $\boldsymbol{X}$ is the reference point in B_0 and $\boldsymbol{\chi}(\boldsymbol{X},t)$ is its current position in B_t; Ω is the Eulerian computational domain.

NOTATION DICTIONARY ACROSS SOURCES

Object	Module 1 (Peskin)	Griffith & Luo (2017)	Module 4 (this course)
material label / reference point	$q \in \mathcal{U}$	$\boldsymbol{s} \in U$	$\boldsymbol{X} \in B_0$
current position	$\boldsymbol{X}(q,t)$	$\boldsymbol{\chi}(\boldsymbol{s},t)$	$\boldsymbol{x} = \boldsymbol{\chi}(\boldsymbol{X},t)$
region occupied by the solid	$\boldsymbol{X}(\mathcal{U},t)$	$\boldsymbol{\chi}(U,t)$	$B_t = \boldsymbol{\chi}(B_0,t)$
Eulerian domain	Ω	Ω	Ω
Lagrangian force density	$\boldsymbol{F}(q,t)$	$\boldsymbol{F}(\boldsymbol{s},t)$	$\boldsymbol{\mathcal{F}}(\boldsymbol{X},t)$
deformation gradient	–	$\partial\boldsymbol{\chi}/\partial\boldsymbol{s}$ (distinct typeface)	$\boldsymbol{F} = \nabla_X \boldsymbol{\chi}$

Two points from Griffith & Luo (2017, §2) worth keeping in mind: the reference coordinates need *not* be the initial positions ($\boldsymbol{\chi}(\boldsymbol{X},0) \neq \boldsymbol{X}$ is allowed), and the reference domain need not lie inside Ω — which is why their Fig. 1 draws U outside the box. The calligraphic $\boldsymbol{\mathcal{F}}$ keeps the force density distinct from the deformation gradient $\boldsymbol{F}$.

INTERFACE TREATMENT IN COMMON FSI METHODS

Method	Fluid mesh	Structural mechanics
Body-fitted / **ALE**	Conforms to the interface. Mesh motion or remeshing may be needed	Standard finite elements and sharp boundary conditions
Classical IB	Fixed Cartesian grid with regularized interface coupling	Lagrangian fibers, springs, or continuum energy models
IBFE	Fixed Cartesian grid with regularized interface coupling	Standard finite elements with weak-form stress divergence

> IBFE couples sophisticated solid mechanics to a fixed, nonconforming fluid grid while retaining a regularized interface.

THE HYBRID DISCRETIZATION

Eulerian fluid solver + Lagrangian FE solid

Finite differences or finite volumes

- incompressible momentum equation
- pressure projection or saddle-point solve
- Cartesian-grid adaptivity when needed

Finite elements

- large-deformation kinematics
- anisotropic or nonlinear material response
- consistent weak-form elastic forces

Regularized delta kernels connect the two discretizations.

ONE VELOCITY FIELD FOR THE COUPLED SYSTEM

The original IBFE formulation uses a composite Eulerian velocity field $\boldsymbol{u}(\boldsymbol{x}, t)$ throughout Ω.

The total Cauchy stress is

$$\boldsymbol{\sigma}(\boldsymbol{x}, t) = \begin{cases} \boldsymbol{\sigma}^f(\boldsymbol{x}, t) + \boldsymbol{\sigma}^e(\boldsymbol{x}, t), & \boldsymbol{x} \in B_t, \\ \boldsymbol{\sigma}^f(\boldsymbol{x}, t), & \boldsymbol{x} \notin B_t. \end{cases}$$

> The background viscous stress acts everywhere. The immersed solid contributes an additional elastic stress inside B_t.

For a Newtonian background fluid,

$$\boldsymbol{\sigma}^f = -p\boldsymbol{I} + \mu\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T\right).$$

Scope of Griffith–Luo (2017): common density and background viscosity. Later extensions treat material contrasts.

CONTINUUM EQUATIONS

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t}+\boldsymbol{u}\cdot\nabla\boldsymbol{u}\right)=-\nabla p+\mu\nabla^2\boldsymbol{u}+\boldsymbol{f}^e,$$

$$\nabla\cdot\boldsymbol{u}=0,$$

$$\frac{\partial \boldsymbol{\chi}}{\partial t}(\boldsymbol{X},t)=\boldsymbol{u}(\boldsymbol{\chi}(\boldsymbol{X},t),t).$$

Three coupled ingredients

- incompressible momentum balance
- Eulerian elastic force $\boldsymbol{f}^e$
- material motion with the local fluid velocity

At the continuum level, the last equation imposes both no slip and no penetration on the immersed material.

KINEMATIC COUPLING AS AN INTEGRAL TRANSFORM

$$\frac{\partial \boldsymbol{\chi}}{\partial t}(\boldsymbol{X}, t) = \int_{\Omega} \boldsymbol{u}(\boldsymbol{x}, t)\, \delta(\boldsymbol{x} - \boldsymbol{\chi}(\boldsymbol{X}, t))\; d\boldsymbol{x}.$$

Continuum identity

The Dirac distribution evaluates the Eulerian velocity at the current material position.

Discrete approximation

A regularized kernel δ_h averages velocity over several grid cells. No slip is therefore enforced in a regularized sense.

> The kernel width controls the numerical interface thickness. It does not represent a physical coating around the solid.

STRESS AND TRACTION IN TWO CONFIGURATIONS

Current configuration

$$\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n}$$

The Cauchy traction is force per unit current area. Its normal and tangential components describe pressure-like and shear loading.

Reference configuration

$$\boldsymbol{t}_0 = \boldsymbol{P}^e\boldsymbol{N}$$

The first Piola–Kirchhoff traction $\boldsymbol{t}_0$ is force per unit reference area. It is the natural surface quantity for a Lagrangian FE weak form.

(we write $\boldsymbol{P}^e$ to stress that it is the elastic stress; later decks drop the superscript)

> The fluid traction on a physical interface and the elastic transmission force used in the IB equations are related by momentum balance, but they are not interchangeable labels.

WHY IBFE USES THE FIRST PIOLA–KIRCHHOFF STRESS

$$\boldsymbol{F} = \nabla_X \boldsymbol{\chi}, \qquad J = \det \boldsymbol{F},$$

$$\boxed{\boldsymbol{P}^e(\boldsymbol{X}, t) = J\, \boldsymbol{\sigma}^e(\boldsymbol{\chi}(\boldsymbol{X}, t), t)\, \boldsymbol{F}^{-T}}$$

- $\boldsymbol{F}$ maps material line elements to the current configuration.
- J gives the local volume ratio.
- $\boldsymbol{P}^e$ need not be symmetric because it links reference normals to current forces.
- Symmetry of $\boldsymbol{\sigma}^e$ appears instead as $\boldsymbol{P}^e \boldsymbol{F}^T = \boldsymbol{F}(\boldsymbol{P}^e)^T$.

> The Piola transformation makes the reference-domain and current-domain traction resultants agree.

Griffith & Luo (2017) define $\boldsymbol{P}^e$ by $\int_{\partial V} \boldsymbol{P}^e \boldsymbol{N}\, dA = \int_{\boldsymbol{\chi}(\partial V, t)} \boldsymbol{\sigma}^e \boldsymbol{n}\, da$ for every subregion V (their Eq. (3)); Nanson's formula $\boldsymbol{n}\, da = J \boldsymbol{F}^{-T} \boldsymbol{N}\, dA$ turns this into the boxed relation (derived in deck 4-2).

ELASTIC STRESS BECOMES AN EULERIAN FORCE

$$\boldsymbol{f}^e(\boldsymbol{x},t) = \int_{B_0} \mathrm{Div}_X\, \boldsymbol{P}^e(\boldsymbol{X},t)\, \delta(\boldsymbol{x} - \boldsymbol{\chi}(\boldsymbol{X},t))\; dV - \int_{\partial B_0} \boldsymbol{P}^e(\boldsymbol{X},t)\boldsymbol{N}\, \delta(\boldsymbol{x} - \boldsymbol{\chi}(\boldsymbol{X},t))\; dA.$$

Internal force

$\mathrm{Div}_X\, \boldsymbol{P}^e$ acts throughout the reference solid and has units of force per reference volume.

Transmission force

$-\boldsymbol{P}^e\boldsymbol{N}$ acts on ∂B_0 and has units of force per reference area. It produces interfacial stress jumps.

Source: Griffith and Luo (2017), Eq. (6) for the boxed force equation (strong form, Eqs. (4)–(7)).

THE UNIFIED WEAK FORCE

Rather than differentiate a piecewise FE stress, define a Lagrangian force density $\mathcal{F}$ by

$$\boxed{\int_{B_0} \boldsymbol{\mathcal{F}} \cdot \boldsymbol{V}\, d\boldsymbol{X} = -\int_{B_0} \boldsymbol{P}^e : \nabla_X \boldsymbol{V}\, d\boldsymbol{X} \qquad \text{for every FE test function } \boldsymbol{V}.}$$

Then spread the result to the fluid:

$$\boldsymbol{f}^e(\boldsymbol{x}, t) = \int_{B_0} \boldsymbol{\mathcal{F}}(\boldsymbol{X}, t)\, \delta(\boldsymbol{x} - \boldsymbol{\chi}(\boldsymbol{X}, t))\; d\boldsymbol{X}.$$

> The test function does not vanish on ∂B_0. Consequently, this one weak equation contains both the internal and boundary transmission forces.

UNIFIED AND PARTITIONED WEAK FORMULATIONS

Unified form

- One volumetric weak force represents internal and boundary terms.
- Adjoint coupling gives an exact discrete interaction-power identity.
- The boundary contribution enters through the volume representation.

Partitioned form

- Internal force and $\boldsymbol{T} = -\boldsymbol{P}^e\boldsymbol{N}$ are spread separately (the reference traction is $\boldsymbol{t}_0 = \boldsymbol{P}^e\boldsymbol{N}$; the transmission force is its negative).
- Coarse solid meshes can give better pressure accuracy and volume conservation.
- The same discrete power identity does not hold.

> The two forms are equivalent before discretization, but their numerical solutions need not coincide.

Source: Griffith and Luo (2017), Sections 2.2 and 7.

FINITE ELEMENTS CARRY THE SOLID KINEMATICS

$$\boldsymbol{\chi}_h(\boldsymbol{X},t) = \sum_{a=1}^{M} \boldsymbol{\chi}_a(t)\,\phi_a(\boldsymbol{X}),$$

$$\boldsymbol{F}_h(\boldsymbol{X},t) = \nabla_X \boldsymbol{\chi}_h = \sum_{a=1}^{M} \boldsymbol{\chi}_a(t) \otimes \nabla_X \phi_a.$$

- Nodal positions define a continuous configuration.
- C^0 elements generally give elementwise-discontinuous $\boldsymbol{F}_h$ and $\boldsymbol{P}_h^e$.
- The weak form avoids differentiating $\boldsymbol{P}_h^e$ across element boundaries.

For basis function ϕ_b, the unified force load is

$$B_b = -\int_{B_0} \boldsymbol{P}_h^e : \nabla_X \phi_b \, d\boldsymbol{X}, \qquad \boldsymbol{M}[\boldsymbol{\mathcal{F}}] = [\boldsymbol{B}].$$

QUADRATURE POINTS MEDIATE FORCE TRANSFER

For each finite element B_0^e, evaluate the FE force density at quadrature points $\boldsymbol{X}_Q^e$:

$$\boldsymbol{f}_h(\boldsymbol{x},t) = \sum_{B_0^e} \sum_{Q=1}^{N_e} \boldsymbol{\mathcal{F}}_h(\boldsymbol{X}_Q^e,t)\, \delta_h\big(\boldsymbol{x} - \boldsymbol{\chi}_h(\boldsymbol{X}_Q^e,t)\big)\; w_Q^e.$$

Classical nodal coupling

Sparse nodes can leave gaps in kernel support and permit numerical leakage.

IBFE quadrature coupling

The element mesh may be coarser than the fluid grid if the quadrature-point net remains sufficiently dense.

Structural resolution still depends on the FE mesh. Denser quadrature improves coupling resolution, not the constitutive approximation.

In Griffith & Luo (2017, §3.3 and §7) the 2D tests use adaptive Gaussian rules giving on average at least a 3×3 net of quadrature points per Cartesian cell, and leak-free results with Lagrangian meshes at least four times coarser than the Eulerian grid (classical nodal IB needs roughly twice as fine). Here w_Q^e are the quadrature weights and N_e the number of points in element B_0^e.

ADJOINT COUPLING PRESERVES INTERACTION POWER

Write the discrete force-spreading and velocity-restriction operations as

$$\boldsymbol{f}_h = \mathcal{S}[\boldsymbol{\chi}_h]\boldsymbol{\mathcal{F}}_h, \qquad \frac{d\boldsymbol{\chi}_h}{dt} = \mathcal{R}[\boldsymbol{\chi}_h]\boldsymbol{u}_h.$$

Choose $\mathcal{R} = \mathcal{S}^*$ with respect to the Eulerian and FE inner products. Then

$$\boxed{\left(\boldsymbol{\mathcal{F}}_h, \frac{d\boldsymbol{\chi}_h}{dt}\right)_{B_0} = (\boldsymbol{f}_h, \boldsymbol{u}_h)_\Omega .}$$

> The Eulerian and Lagrangian descriptions assign the same instantaneous power to the coupling force. This property alone does not guarantee stability of the full time integrator.

In matrix form (2D grid spacing h, FE mass matrix $\boldsymbol{M}$): $[\mathcal{R}] = \boldsymbol{M}^{-1}[\mathcal{S}]^T h^2$, so $d\boldsymbol{\chi}_h/dt$ is the L^2 projection of the kernel-interpolated velocity onto the FE space. Source: Griffith & Luo (2017), Eqs. (35)–(40). The partitioned form does not satisfy this identity (§7 there).

ONE IBFE TIME STEP

1. Restrict the Eulerian velocity to FE degrees of freedom.
2. Update the current configuration $\boldsymbol{\chi}_h$.
3. Evaluate $\boldsymbol{F}_h$, J_h, and the constitutive stress $\boldsymbol{P}_h^e$.
4. Assemble the Lagrangian weak force coefficients.
5. Spread quadrature-point forces to the Cartesian grid and advance the fluid.

> Partitioned, semi-implicit, and fully implicit algorithms reorder or couple these operations differently. The transfer operators and variational solid mechanics remain the organizing ideas.

CONSTITUTIVE MODELS IN THE FE SOLID

Hyperelastic example

$$\boldsymbol{P}^{e}_{\text{pas}} = \frac{\partial W}{\partial \boldsymbol{F}}$$

The strain-energy density W may represent isotropic, fiber-reinforced, or nearly incompressible response.

More general stress laws

The numerical framework needs a routine that evaluates a Lagrangian stress. It can also accommodate history-dependent or viscoelastic response when the required internal variables are evolved.

> Griffith and Luo use hyperelastic examples for simplicity, but the method itself does not require an explicit strain-energy functional.

ACTIVE TENSION BELONGS IN THE STRESS MODEL

For a passive material with an active fiber response, decompose

$$\boldsymbol{P}^e = \boldsymbol{P}^e_{\text{pas}} + \boldsymbol{P}^e_{\text{act}}.$$

A common construction

- $\boldsymbol{a} = \boldsymbol{F}\boldsymbol{f}_0$ (mapped fiber; $\boldsymbol{f}_0$ a unit reference direction). Note $\boldsymbol{a}$ is *not* normalized.
- prescribe an active Cauchy stress: $\boldsymbol{\sigma}^e_{\text{act}} = T\,\boldsymbol{a} \otimes \boldsymbol{a}$.
- map it to the reference configuration: $\boldsymbol{P}^e_{\text{act}} = J\boldsymbol{\sigma}^e_{\text{act}}\boldsymbol{F}^{-T} = J\,T\,\boldsymbol{a} \otimes \boldsymbol{f}_0$.

Model choices

- T may depend on activation, stretch, and stretch rate.
- Other conventions normalize $\boldsymbol{a}$ or absorb stretch factors into T; see deck 4-3.

> A stress-based activation law makes material-frame invariance and local momentum balance explicit.

ACTIVE TENSION: CHECKING THE PULL-BACK

The only identity needed is $(\boldsymbol{a}\otimes\boldsymbol{b})\boldsymbol{C}=\boldsymbol{a}\otimes(\boldsymbol{C}^T\boldsymbol{b})$. With $\boldsymbol{a}=\boldsymbol{F}\boldsymbol{f}_0$,

$$\boldsymbol{P}^e_{\text{act}} = J\,T\,(\boldsymbol{a}\otimes\boldsymbol{a})\boldsymbol{F}^{-T} = J\,T\,\boldsymbol{a}\otimes(\boldsymbol{F}^{-1}\boldsymbol{a}) = J\,T\,\boldsymbol{a}\otimes\boldsymbol{f}_0 = J\,T\,\boldsymbol{F}\boldsymbol{f}_0\boldsymbol{f}_0^T.$$

Stretch factors

Because $|\boldsymbol{a}|=\lambda_f$ (the fiber stretch), $\boldsymbol{\sigma}^e_{\text{act}} = T\lambda_f^2\,\hat{\boldsymbol{a}}\otimes\hat{\boldsymbol{a}}$ with $\hat{\boldsymbol{a}}=\boldsymbol{a}/\lambda_f$. The physical Cauchy fiber stress is therefore $T\lambda_f^2$, not T.

Unit-direction convention

If instead $\boldsymbol{\sigma}^e_{\text{act}} = \widehat{T}\,\hat{\boldsymbol{a}}\otimes\hat{\boldsymbol{a}}$, then $\boldsymbol{P}^e_{\text{act}} = (J\widehat{T}/\lambda_f^2)\,\boldsymbol{a}\otimes\boldsymbol{f}_0$. The two scalars are related by $\widehat{T}=T\lambda_f^2$.

> Symbol check: the scalar active tension T is unrelated to the bold transmission force $\boldsymbol{T}=-\boldsymbol{P}^e\boldsymbol{N}$ and to the final time in $[0,T]$. Always report which convention calibrates T.

INCOMPRESSIBILITY AND STRUCTURAL VOLUME

At the continuum level, material motion with a divergence-free velocity gives

$$\frac{dJ}{dt} = J \nabla \cdot \boldsymbol{u}(\boldsymbol{\chi}(\boldsymbol{X}, t), t) = 0.$$

Continuum consequence

An initially volume-preserving deformation remains volume preserving under the exact incompressible velocity field.

Discrete reality

Kernel regularization, velocity projection, time integration, and under-resolution can introduce volume drift.

> Mesh refinement, compatible coupling, appropriate quadrature, and volumetric stabilization can reduce drift. None replaces a convergence study.

WHAT IBFE DOES ESPECIALLY WELL

- Handles large translations, rotations, and deformations without remeshing the fluid grid.
- Reuses standard finite element ideas for nonlinear and anisotropic solid mechanics.
- Separates structural mesh spacing from coupling-point density through element quadrature.
- Fits naturally with Cartesian-grid solvers and block-structured adaptive mesh refinement.

> The main benefit is a flexible division of labor: the Eulerian discretization handles incompressible flow, while the FE discretization handles material mechanics.

PERSISTENT NUMERICAL CHALLENGES

- The regularized kernel spreads the interface over several grid cells, which limits sharp stress and pressure resolution.
- Stiff elastic response can impose severe time-step restrictions in explicit or loosely coupled schemes.
- Contact, self-contact, tearing, and topology change require additional models or algorithms.
- Density and viscosity contrast require extensions beyond the simplest one-field formulation.
- Accurate results still require adequate fluid resolution, FE resolution, and coupling quadrature.

> IBFE improves geometric flexibility, but sharp interfaces and stiff material laws remain costly.

WHEN IBFE IS A GOOD MATCH

Strong fit

- large solid motion or deformation
- detailed nonlinear constitutive behavior
- fixed material topology
- tolerance for a regularized interface

Consider another formulation when

- pointwise surface stresses or thin boundary layers dominate the objective
- rupture or changing topology is central
- a reliable body-fitted mesh remains inexpensive to maintain

> Method choice should follow the required observables: bulk motion and deformation, or sharply resolved interfacial fields.

REPRESENTATIVE IBFE PROBLEMS

Benchmarks in Griffith–Luo

- anisotropic elastic shells
- an elastic disc in a lid-driven cavity
- pressure-driven collapsible-channel flow
- the Turek–Hron cylinder-and-flap benchmark

Broader application areas

- heart valves and cardiovascular flow
- deformable biological tissues
- flexible structures in viscous flow
- actively contracting materials

See Griffith and Luo (2017) and Griffith and Patankar, *Annu. Rev. Fluid Mech.* 2020.
Benchmarks: Griffith & Luo (2017) §6.1 (anisotropic and orthotropic thick elliptical shells), §6.2 (soft neo-Hookean disc in a lid-driven cavity), §6.3 (collapsible channel), §6.4 (Turek–Hron). Application areas: Griffith & Patankar (2020) §5 (aquatic locomotion, esophageal transport, cardiovascular dynamics).

RESOURCES

- Griffith and Luo (2017): hybrid finite difference / finite element IB method
- Griffith and Patankar (2020): immersed methods for FSI
- IBAMR software and documentation
- Boyce Griffith presentation on IBFE
- Solid mechanics and IBFE tutorial by Matea
- AIM/MICDE seminar: immersed methods for FSI

THREE IDEAS TO CARRY FORWARD

1. Mechanics: finite elements provide the deformation map, stress, and weak elastic force.
2. Coupling: regularized integral operators exchange force and velocity between nonconforming grids.
3. Numerics: quadrature density, adjointness, and resolution determine whether the hybrid method realizes its continuum properties.

Questions?

REFERENCES

- B. E. Griffith and X. Luo, Hybrid finite difference/finite element immersed boundary method, *Int. J. Numer. Methods Biomed. Eng.* 33 (2017) e2888. doi:10.1002/cnm.2888. (Primary source for this deck: Eqs. (1)–(40), §§2–3, §6, §7; Fig. 1 reproduced on slide 4.)
- D. Boffi, L. Gastaldi, L. Heltai, and C. S. Peskin, On the hyper-elastic formulation of the immersed boundary method, *Comput. Methods Appl. Mech. Engrg.* 197 (2008) 2210–2231. doi:10.1016/j.cma.2007.09.015. (Continuous formulation used by Griffith & Luo.)
- B. E. Griffith and N. A. Patankar, Immersed methods for fluid–structure interaction, *Annu. Rev. Fluid Mech.* 52 (2020) 421–448. doi:10.1146/annurev-fluid-010719-060228.
- C. S. Peskin, The immersed boundary method, *Acta Numerica* 11 (2002) 479–517. doi:10.1017/S0962492902000077. (Module 1 notation; nodal-coupling rule of thumb.)
- A. P. Hoover, B. E. Griffith, and L. A. Miller, Quantifying performance in the medusan mechanospace with an actively swimming three-dimensional jellyfish model, *J. Fluid Mech.* 813 (2017) 1112–1155. doi:10.1017/jfm.2017.3. (Active-stress example developed in deck 4-3.)
- G. A. Holzapfel, *Nonlinear Solid Mechanics: A Continuum Approach for Engineering*, Wiley, 2000. (Piola transformation, Nanson's formula, hyperelasticity.)
- IBAMR: an adaptive and distributed-memory parallel implementation of the immersed boundary method, ibamr.github.io.

Title-slide schematic: instructor's course notes. Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.